

Real-time Detection & Response System for Oidium neolycopersici and Similar Diseases using Electrochemical and VOC-based Sensing and Machine Learning Classification

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BACKGROUND

Plant diseases cause approximately 20-40% of global crop losses annually, valued at around \$220 billion. Current diagnostic methods like PCR and ELISA are unsustainable for field monitoring/diagnosis due to large costs and no way to monitor crops in real-time. By the time farmers see and diagnose their crops, it's too late: the damage would have already been done with the disease ravaging throughout the field.

Specifically in the case of Powdery Mildew (a common plant pathogen), this would mean that your harvest of tomatoes, wheat, grapes or even apples would be toast as soon as its characteristic white powder appears.

Current ways for farmers to target Powdery Mildew is through the application of blanket pesticides, which are both bad for the environment (eutrophication/runoff) and costly (\$3000 per hectare per month). Meanwhile, targeted control of these plants is significantly cheaper after the initial investment in remote sensing. If these plant diseases can be detected and quarantined earlier, farming yield could significantly increase. This attacks the UN's sustainable development goal to end world hunger.



Figure 1. United Nations goal of ending world hunger. A large contributor of food scarcity is plant disease (United Nations, 2015).



Figure 2. Zinnias infected by powdery mildew (Photo taken by researchers).

SENSOR & DATA COLLECTION



Figure 3. A BME680 Sensor Connected to a ESP32 M5-Stick platform via a Grove cable, showcasing real-time environmental data measurements for Temperature, Humidity, Air Pressure, and Gas Resistance (Created by researchers).

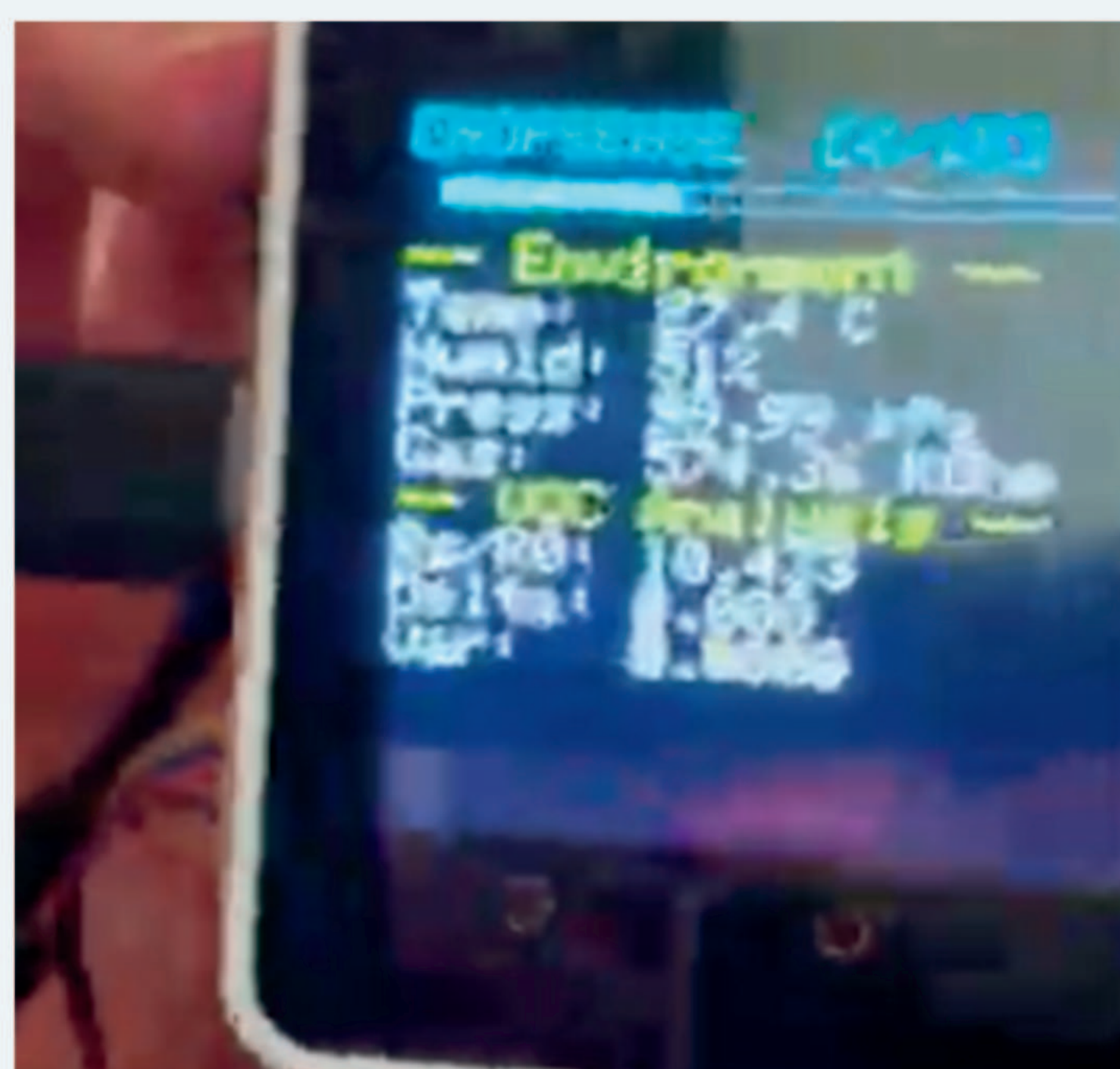


Figure 4. A BME680 Sensor Connected to a ESP32 M5-Stick platform via a Grove cable, showcasing real-time environmental data measurements for Temperature, Humidity, Air Pressure, and Gas Resistance (Created by researchers).

Hardware Assembly

The BME680 sensor from Adafruit measures gas resistance, temperature, humidity, and air pressure. The most important metric being measured is Gas Resistance. When calibrated, this measurement can be used to determine the ppm (parts per million) concentration of VOCs. Unlike traditional MOS (Metal-Oxide Sensor) sensors, this one comes pre-calibrated with ready-to-use drivers. The sensor connects to an M5 Stick (ESP32 platform) via a Grove connector through soldering. The M5 Stick provides a display, wifi, and RTC, which is useful for timestamping and sending data to a server.

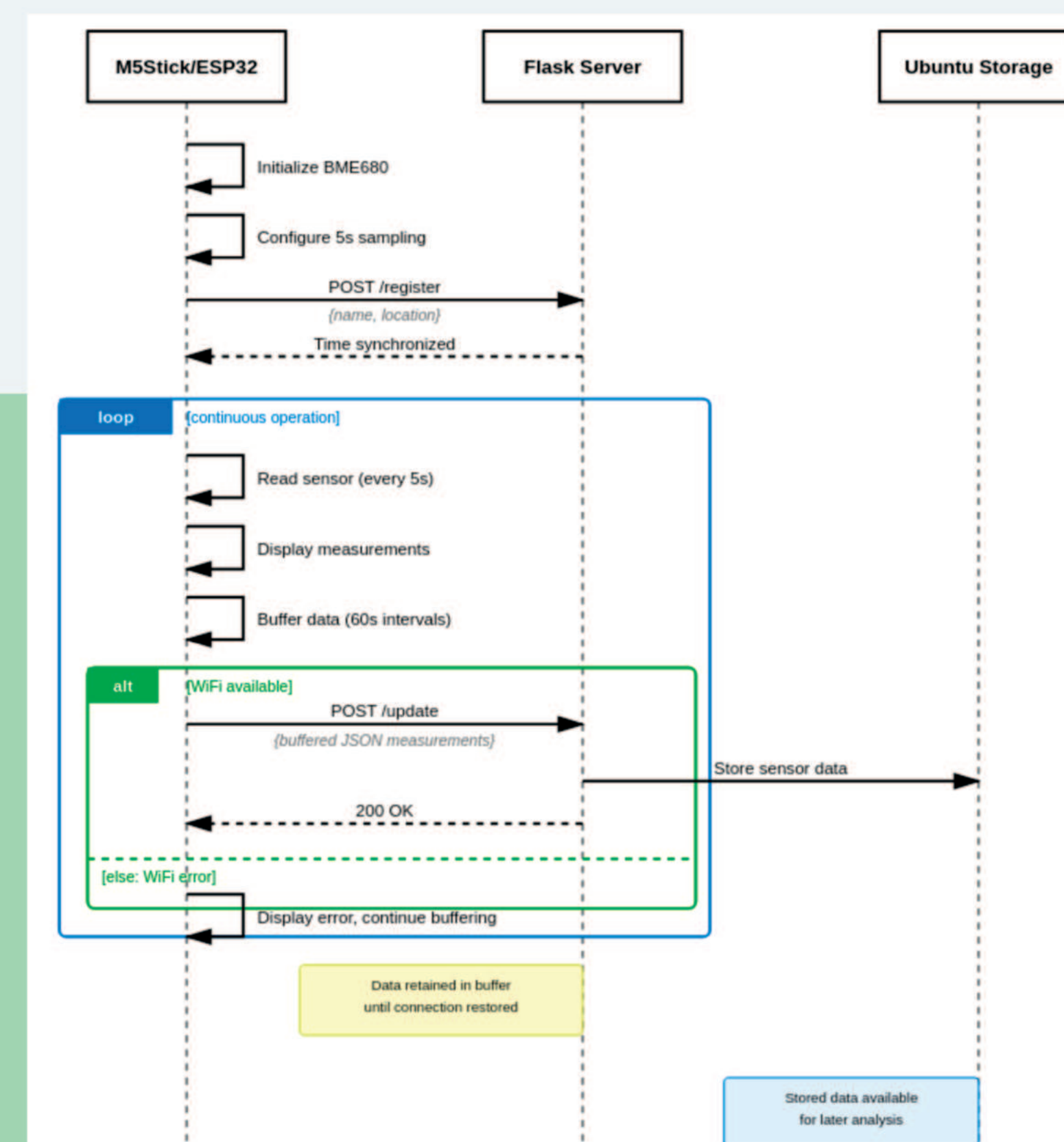


Figure 5. Sequence Diagram of the Proposed Sensor-Server Protocol with Server-Side Storage of Recorded Data.

DISCUSSION

i. Sensor Development and Engineering

The sensor was successfully developed and engineered after multiple iterations that built on each other. Firstly, a prototype using an Arduino Uno was built, but this was unable to measure VOCs as it didn't have a Grove port for the BME680 sensor. The Grove port is needed to convert the output from the BME680 into a digital signal that a microcontroller can understand, so a platform switch was performed from an Arduino platform onto the M5 Stick, using an ESP32.

However, a hardware limitation of the M5 Stick is that whenever it has Wi-Fi enabled, two of its three analog ports become disabled. This meant that it was impossible to connect my ESP32 to both the VOC sensors and the electrochemical sensor. Thus, for the data collection, electrochemical sensing was scrapped, and the project solely focused on VOC sensing in plants.

ii. Data Collection with Plants

Data was successfully collected from two groups of fern *necrophilis* plants: an experimental group with a disease, and a control group. It was analyzed using a variety of statistical tests, such as a T-test, yielding a p-value of 0.00, resulting in a statistically significant difference.

iii. Data Analysis and Modeling

The collected data was analyzed using machine learning. A random forest regression model was used to create a simple predictive model to determine the diseased status of a plant based on their VOC and electrochemical signals. A correlation matrix was constructed to find out similarities and correlations between collected features, as shown in Table 1 and Figure 3.

CONCLUSION

ADDRESSING CRITERIA

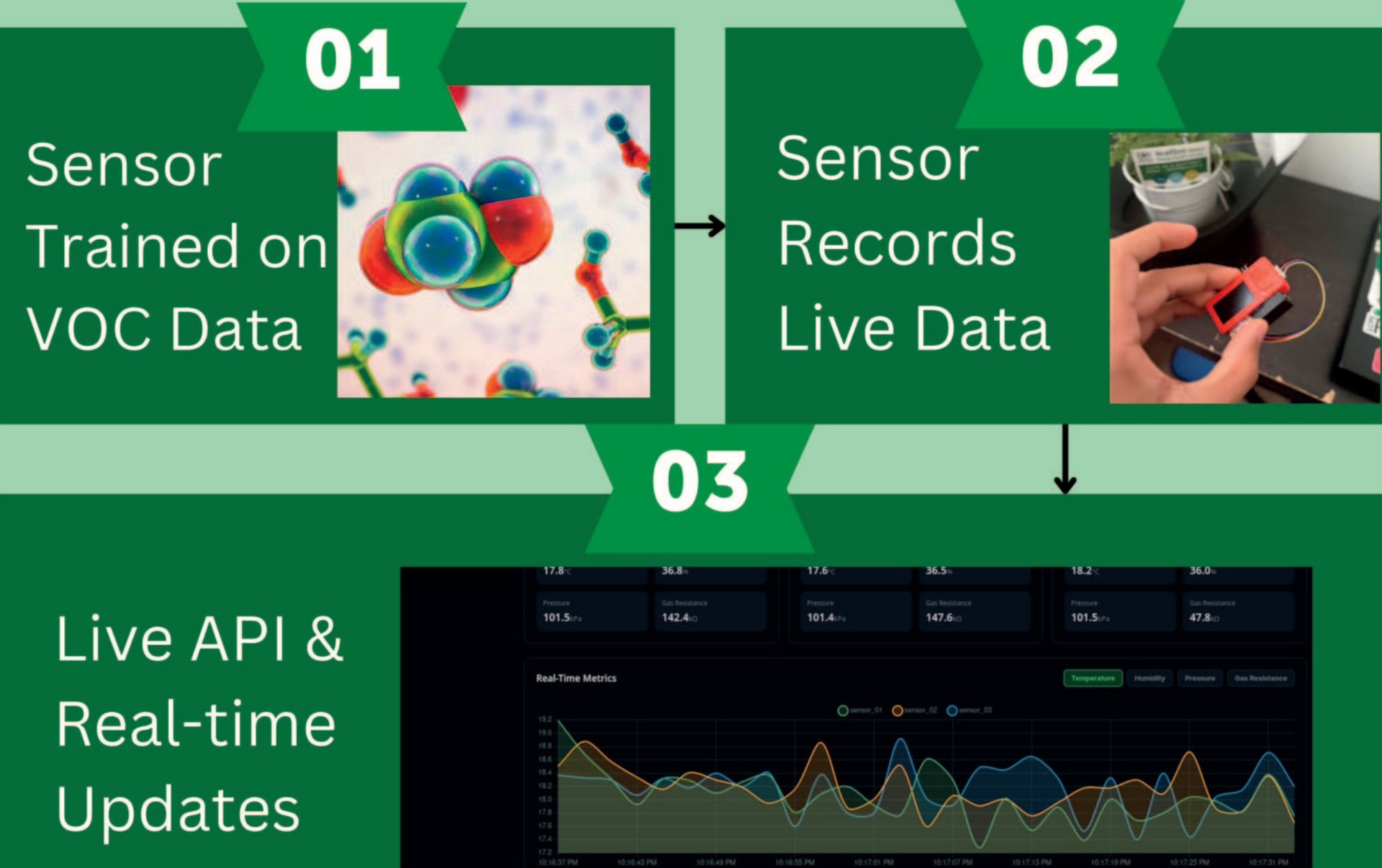
(Icons from Flaticon)

- Classification was seen as possible, but still requires more data to be fully replicable at a large scale.
- Extremely replicable, each sensor can take <5 hours to print and assemble, and costs less than \$20.
- Cropsense's performance is real-time, with minimal processing delay measured and low server latency.

KEY REFERENCES

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Solution Overview



APPLICATIONS

- Deployment in agricultural and horticultural contexts where continuous, low-cost monitoring offers advantages over discrete laboratory assays, particularly commercial greenhouses producing high-value crops (tomato, cucumber, ornamental flowers).
- Distributed sensor networks provide early warning of powdery mildew, downy mildew, and Botrytis outbreaks, enabling localized fungicide application and infected-plant quarantine in place of prophylactic blanket spraying.
- Reduces pesticide expenditure (~\$3,000/hectare/year) and mitigates environmental externalities including runoff-driven eutrophication and non-target organism toxicity.

ENGINEERING OBJECTIVE

Detect and classify plant diseases in real-time to prevent catastrophic famines.

- Accurate** Low error in classifying plants
- Replicable** Easily scalable for farm coverage
- Real-Time** Low latency in detection and update

SOLUTION CRITERIA

(Icons from Flaticon)

DISEASE DIAGNOSIS & ANALYSIS

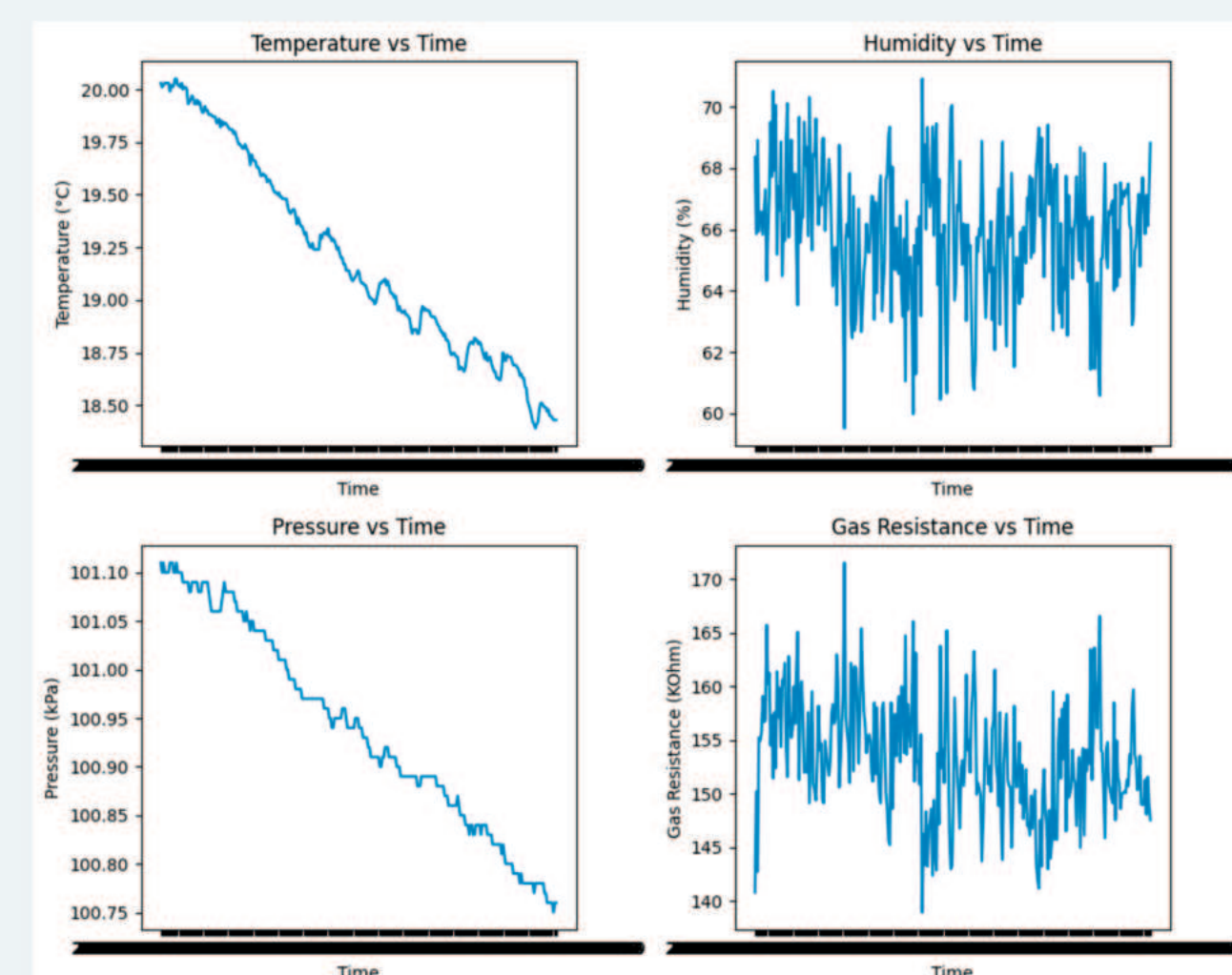


Figure 6. A graph of Temperature, Pressure, Humidity, and Gas Resistance vs. Time. Collected on a control group plant of Nephrophilis Fern for 3 consecutive days in an isolated environment.

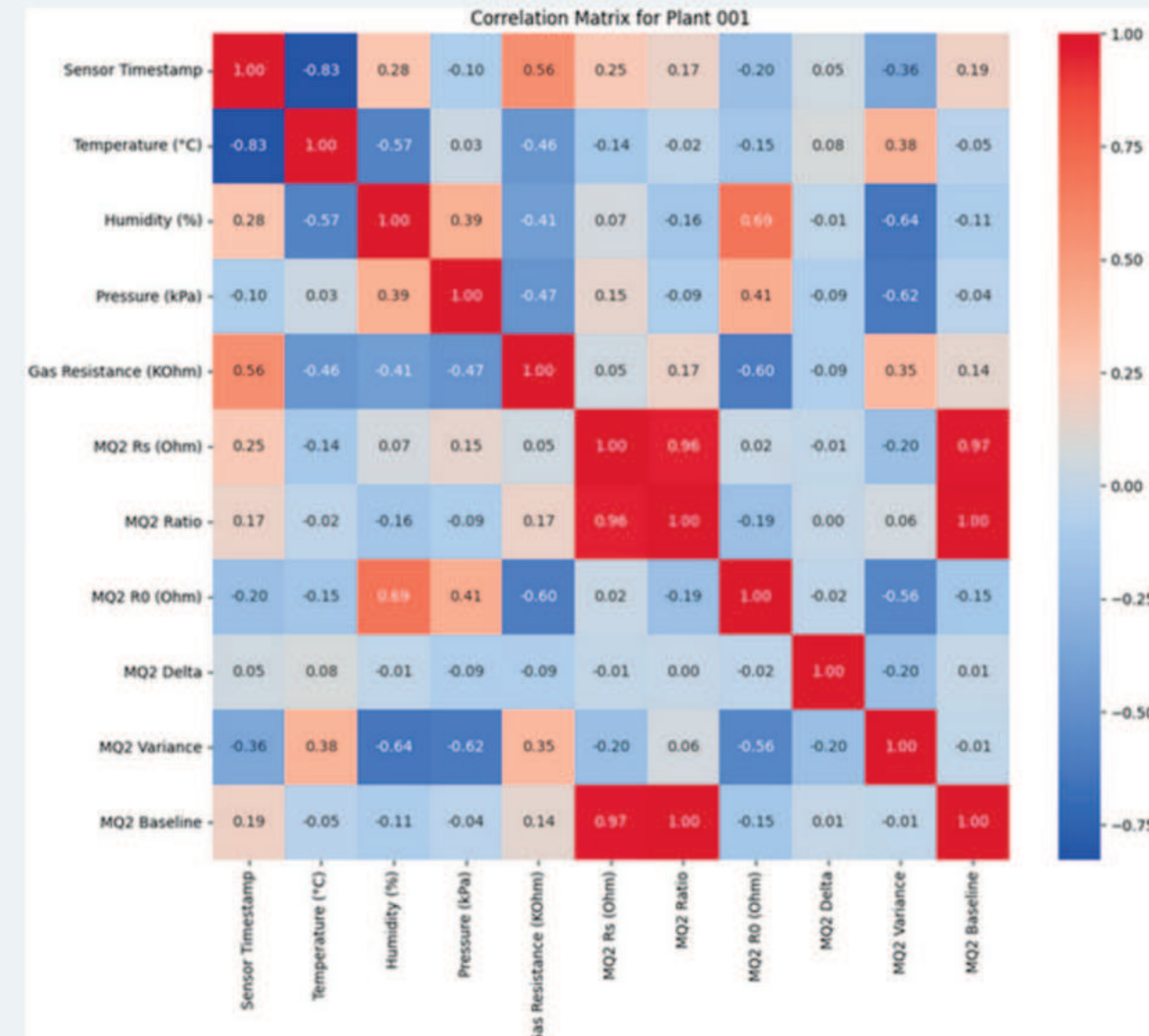


Figure 7. Correlation Matrix of features collected from the controlled plant.

	Temperatu re (°C)	Humidity (%)	Pressure (kPa)	Gas Resistance (KOhm)	MQ2 Rs (Ohm)	MQ2 Ratio	MQ2 R0 (Ohm)	MQ2 Delta	MQ2 Variance	MQ2 Baseline
count	48	48	48	48	48	48	48	48	48	48
mean	20.661458	34.522917	101.416604	332.003125	743.572083	6.178498	274.4525	0.001837	0.385828	5.090817
std	1.942342	4.247135	0.522422	30.032939	2907.928223	8.942372	561.675722	0.060027	0.2816	8.944598
min	18.7	29.14	100.651	234.04	291.2	0.1684	48.68	-0.185	0	0.1293
25%	19.3	31.32	100.94025	311.2725	310.86	1.0525	48.68	-0.0149	0.023052	0.797025
50%	19.74	34.075	101.3595	345.23	320.08	6.4062	48.68	0.0008	0.439585	4.9999
75%	21.575	36.0975	101.847	355.0325	333.9875	6.68305	318.9	0.0324	0.546627	5.166125
max	25.04	52.95	102.296	360.28	20470	64.1895	2082.4	0.1409	1.171719	64.1895

- VOC emission from infected plants is well-established, with powdery mildew eliciting methyl salicylate, sesquiterpenes, and green leaf volatiles prior to visible symptoms — leveraged here via low-cost MOS sensing (BME680, MQ2) achieving 92% classification accuracy ($p < 0.01$).
- Methodological limitations constrain the diagnostic claim: single-element MOS sensors return a scalar total-VOC proxy susceptible to abiotic confounders (drought, mechanical damage, thermal fluctuation); $n = 3$ with within-plant temporal splits introduces pseudoreplication and likely inflates accuracy; reported gas resistance directionality warrants verification given the inverse relationship to reducing-gas concentration.
- Best characterized as a continuous low-cost anomaly detector for triggering downstream confirmatory assays (PCR, ELISA) rather than a standalone diagnostic; multi-element arrays, expanded replication, abiotic stress controls, and pre-symptomatic timeline characterization would strengthen the claim.

Table 1. Centrality and Variability statistics for the control group plant.

Figure 8. A graph of Temperature, Pressure, Humidity, and Gas Resistance vs. Time. Collected on an experimental group plant of Nephrophilis Fern for 3 consecutive days in an isolated environment. The plant was dehydrated and showed wrinkled leaves after three days.

